Lecture 7 - Interpreting, Estimating, and Applying the 1st Derivative and the 2nd Derivative

Garrett Credi

09/29/26

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- Start early! The later you start, the less time you'll have to ask me questions in office hours/via email!
- Review session - I'll hold an Exam 1 review session next Monday (from 6-8?) where you can come and ask any and all questions you have and I'll answer em!

Quiz Time!

Please get into neat rows, and spread out. Only then will I hand out the quiz. As per usual, no cheating (hopefully should go without saying).

Interpreting and Applying the First Derivative

Derivative ReCap

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and determine some information about $f(x)$ from $f'(x)$.

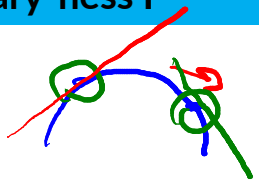
Increasing-ness, Decreasing-ness, and Stationary-ness I

We can use the derivative to determine whether a function is increasing/decreasing/stationary:

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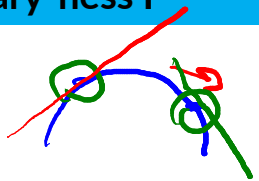
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A function $f(x)$ is increasing around a point $x = a$ if it is increasing in *some* small neighborhood around $x = a$.

Decreasingness is similarly defined.

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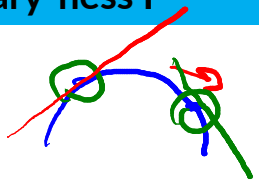
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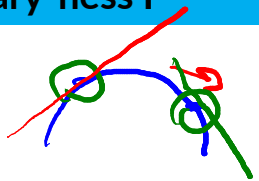
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Increasingness via the Derivative

For a function $f(x)$ which is differentiable at $x = a$, increasingness at a is equivalent to $f'(a) > 0$. Similarly, decreasingness is equivalent to $f'(a) < 0$.

Increasingness II

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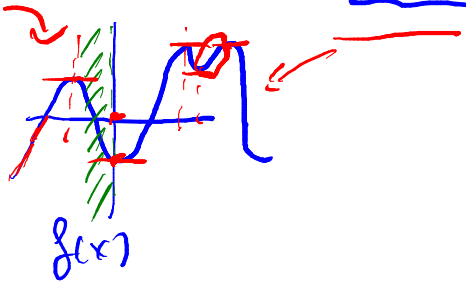
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In terms of instantaneous velocity: positive velocity means increasing position, resp. negative velocity means decreasing position.

Graph Sketching I

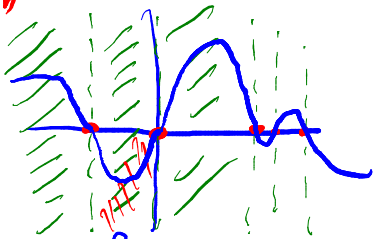
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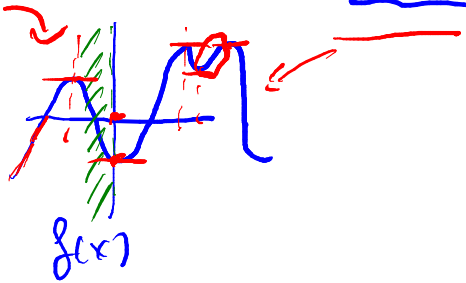
2.) Determine pos/neg

3.) Vaguely draw



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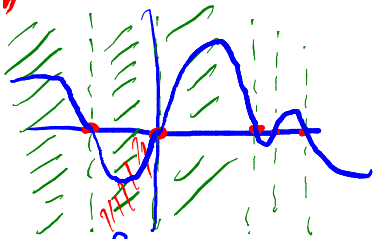
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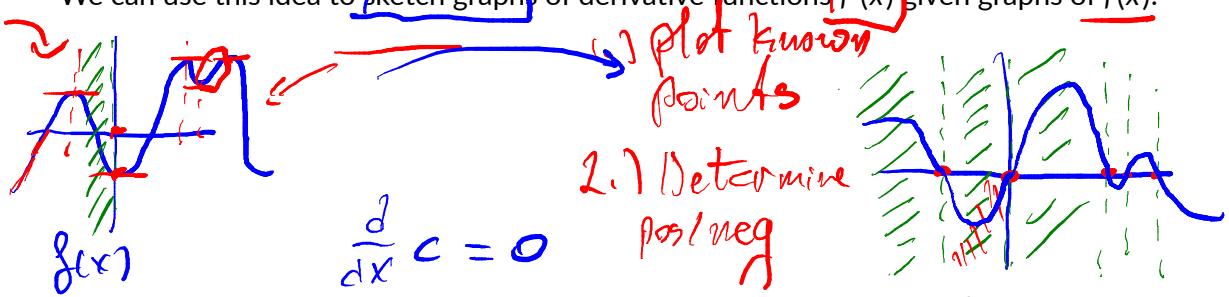
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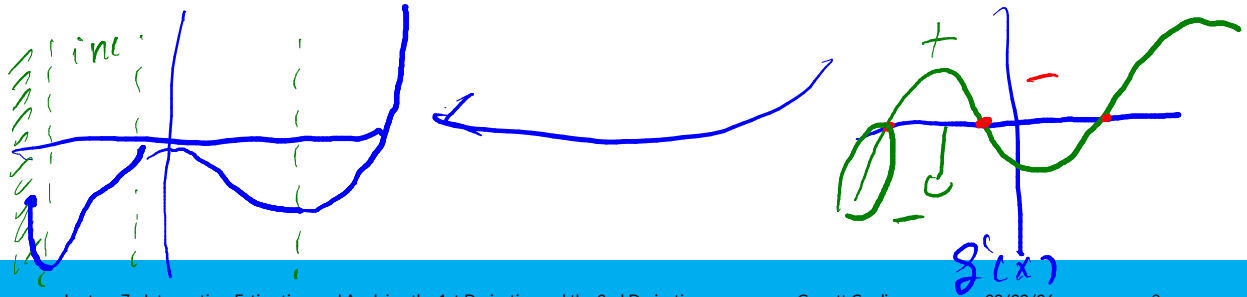


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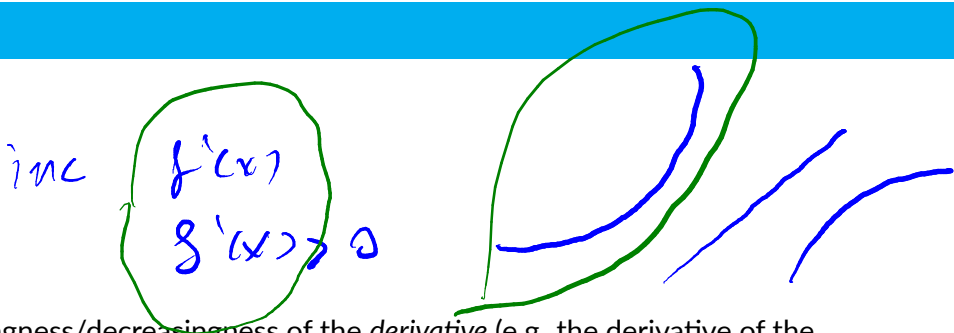
And to sketch graphs of function $f(x)$ given graphs of their derivatives $f'(x)$:



Graph Sketching II

Notice how the increasingness/decreasingness of the *derivative* (e.g. the derivative of the derivative) informed us about the shape of the original function $f(x)$

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Notice how the increasingness/decreasingness of the *derivative* (e.g. the derivative of the derivative) informed us about the shape of the original function $f(x)$ - we'll see later in lecture this is due to the properties of the *second* derivative $f''(x)$.

$$g(x) = x^2 + 2x + 1$$

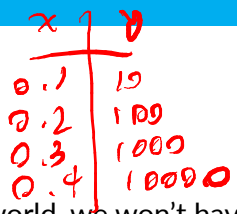
Estimating The Derivative

Estimation I



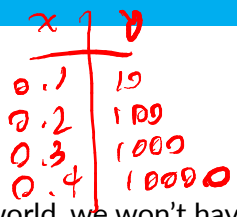
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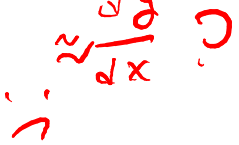
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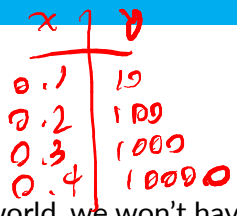
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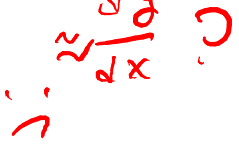




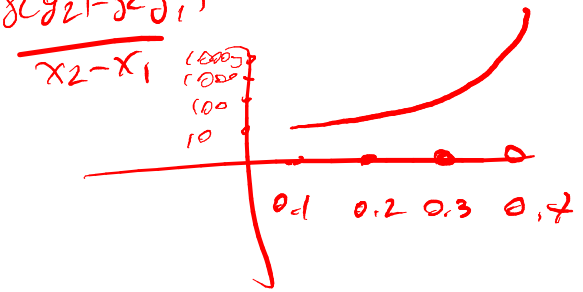
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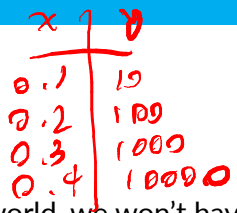
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If we're trying to estimate:

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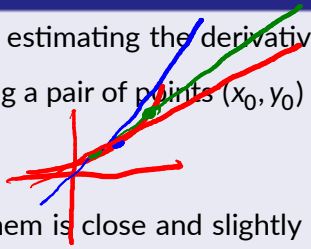
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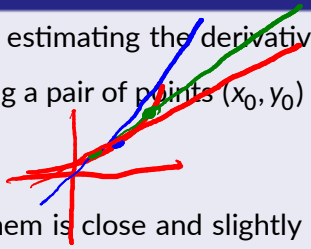
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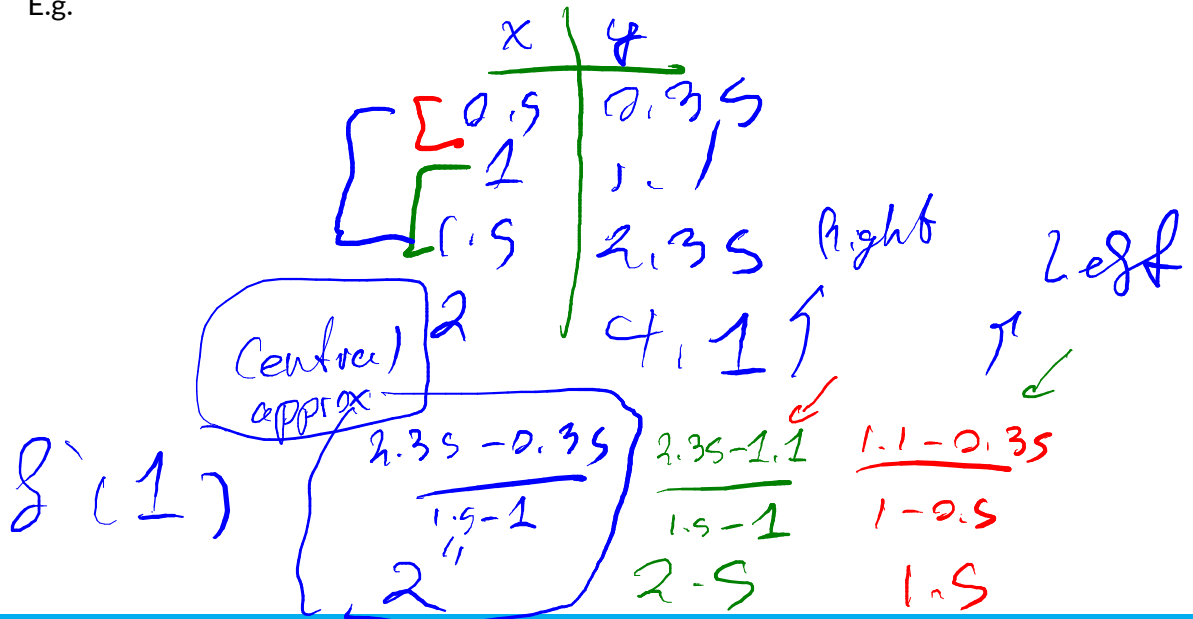
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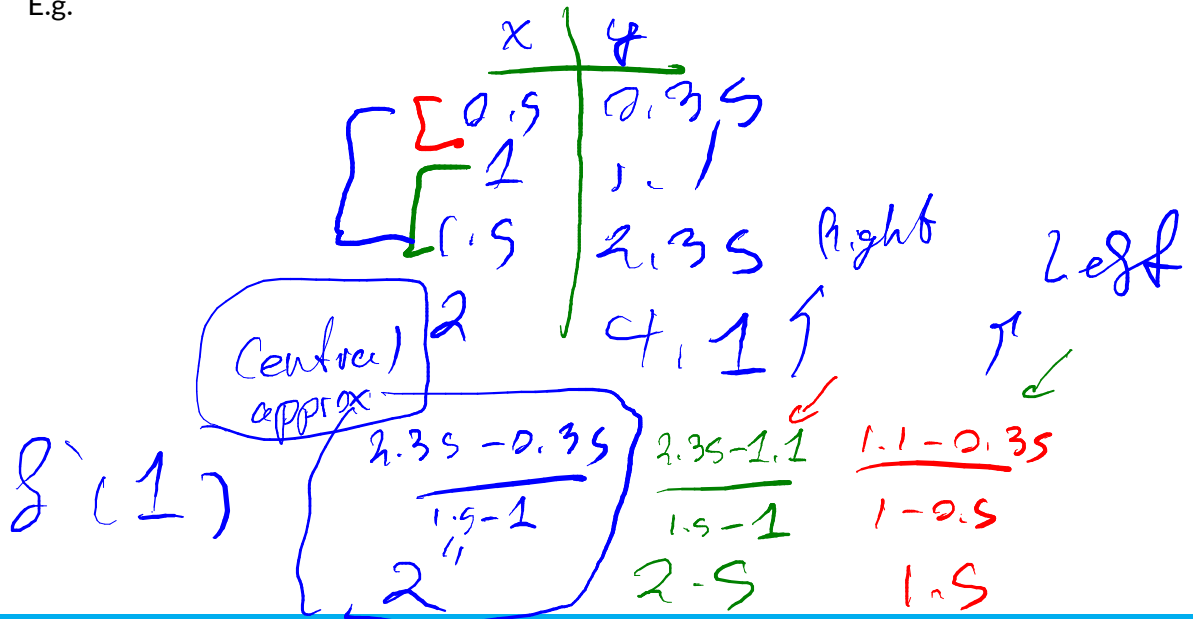
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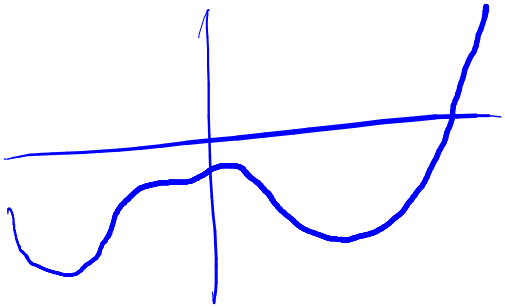
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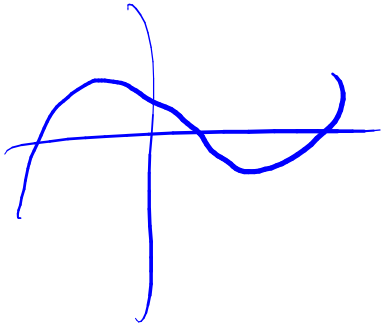


Second and Higher Derivatives

Graph Sketching Redux

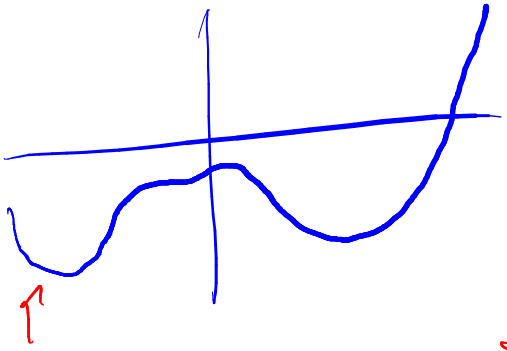
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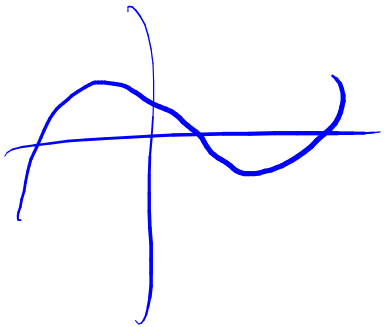


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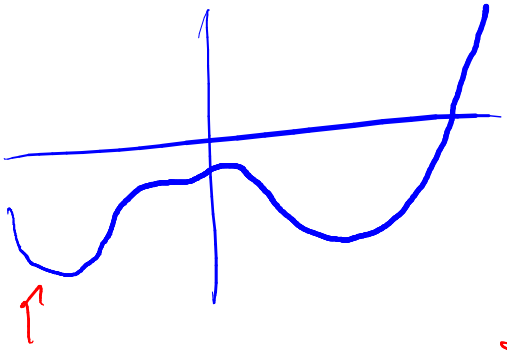
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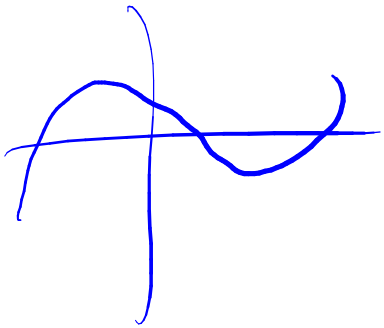


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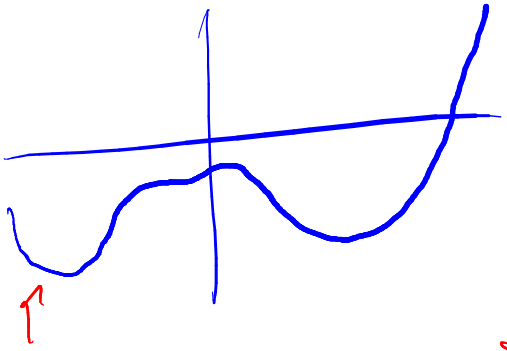


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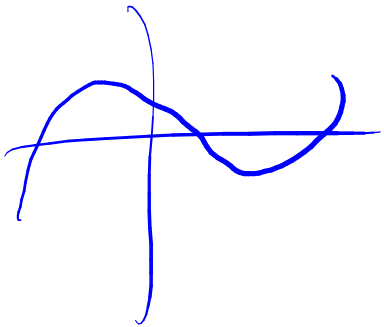
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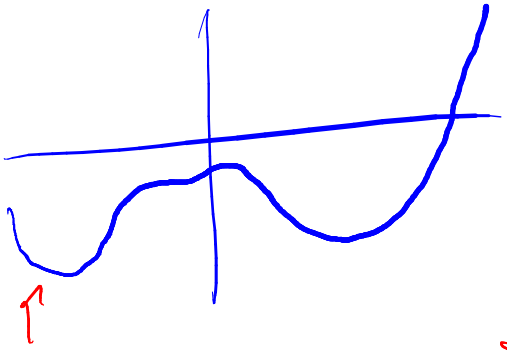
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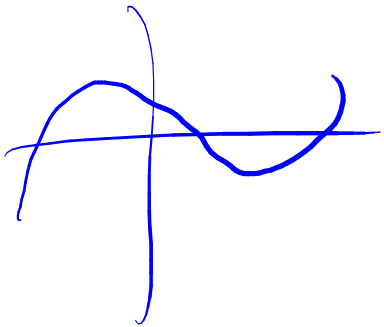
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Concavity in terms of the Second Derivative

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E.g.

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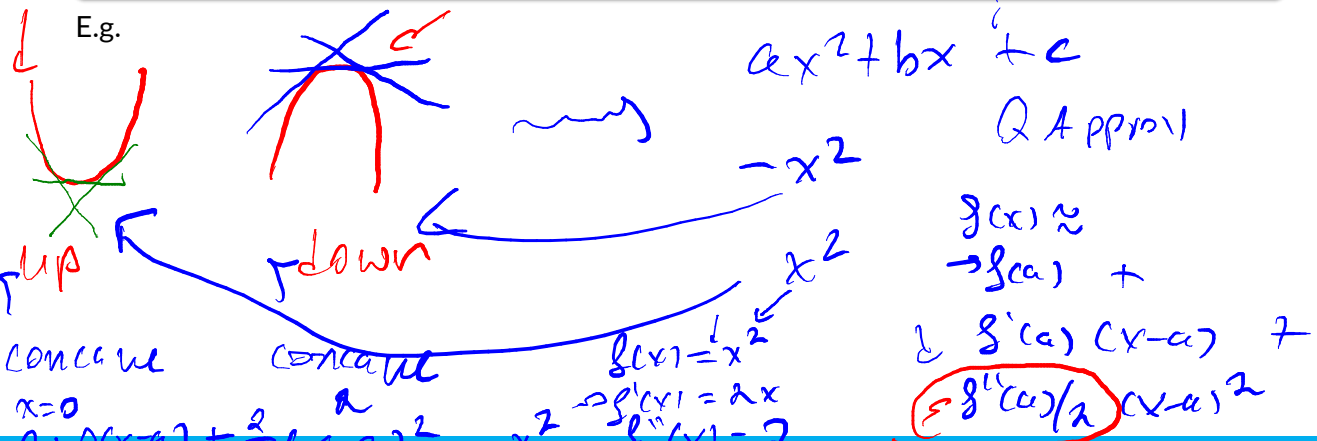
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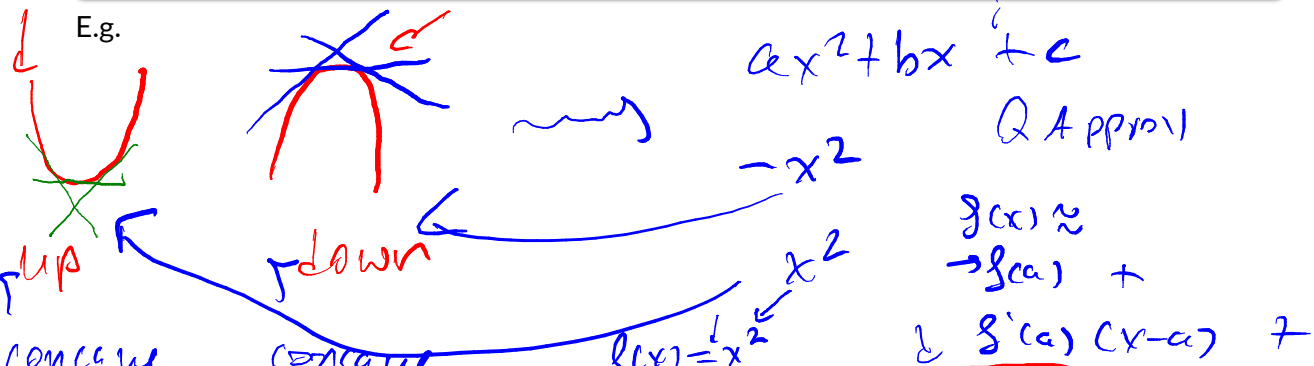
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Classifying Shapes of Functions via the First and Second Derivatives

We generically have four types of shapes a function can take

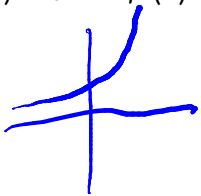
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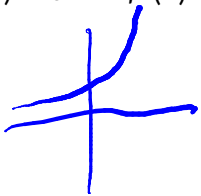
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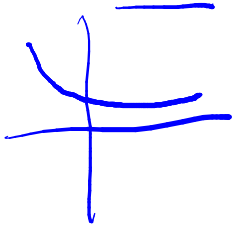
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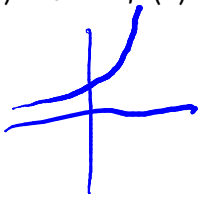
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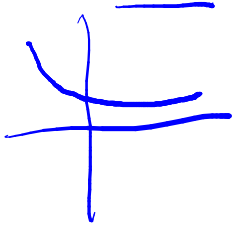
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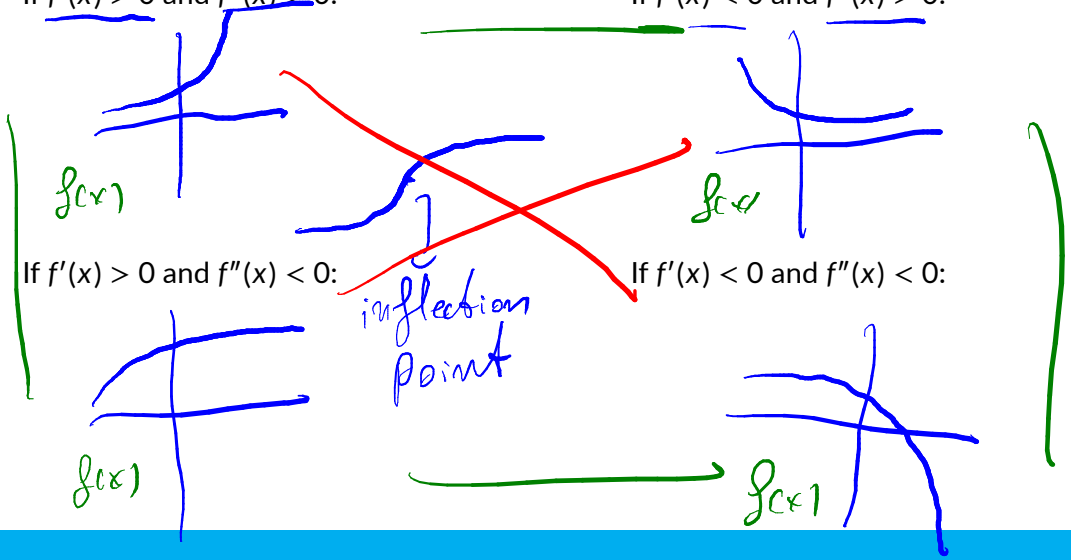


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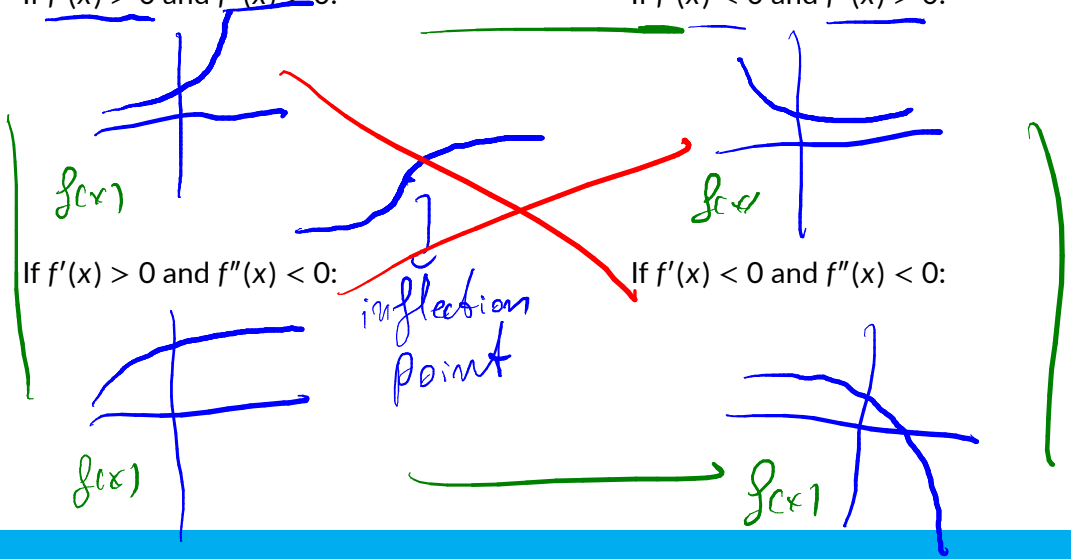
If $f'(x) < 0$ and $f''(x) < 0$:

Classifying Shapes of Functions via the First and Second Derivatives

We generically have four types of shapes a function can take given information about it's first and second derivatives:

If $f'(x) > 0$ and $f''(x) > 0$:

If $f'(x) < 0$ and $f''(x) > 0$:



If $f'(x) > 0$ and $f''(x) < 0$:

If $f'(x) < 0$ and $f''(x) < 0$:

Worksheet Time!

Let’s get into groups and work on the worksheet!

End of Class

That’s all folks!